Modular Tensor Category: with application to CFT

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Abstract

This is a note on the topic of Modular Tensor Categories. It is mainly based on the discussion in [1]. We assume readers have basic understandings of 2D CFT. YL: [This is a note on Tensor Categories, Yet its better to stop cause I have a better reference, but the finished content is still useful.]

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1 Algebras and Representations

This section we introduce some concepts in Algebra and Representation Theory. We will see that these concepts are very useful in characterizing structures in CFT and they naturally forms a category structure, which is the main topic of the next section.

1.1 Intertwiner and Representation

Let G be a group and let (V, ρ), (W, σ) be two representations of G. A **intertwiner** (or **morphism of representations**) is a linear map

$$\varphi : V \to W \tag{1}$$

such that for all $g \in G$,

$$\varphi \circ \rho(g) = \sigma(g) \circ \varphi, \tag{2}$$

i.e., the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho(g)} & V \\ \varphi\downarrow & & \downarrow\varphi \\ W & \xrightarrow[\sigma(g)]{} & W \end{array} \tag{3}$$

The set of all intertwiners from (V,ρ) to (W,σ) is denoted $\mathrm{Hom}_G(V,W)$. If φ is an isomorphism, the two representations are said to be **equivalent** (or **isomorphic**), written $(V,\rho)\simeq(W,\sigma)$.

Theorem 1.1 **Schur's Lemma**

Let (V,ρ) and (W,σ) be irreducible representations of G, and let $\varphi\in\mathrm{Hom}_G(V,W)$.
1. If $(V,\rho)\ncong(W,\sigma)$, then $\varphi=0$.
2. If $V=W$ and G acts over $\mathbb{C}$, then $\varphi=\lambda\mathrm{id}_V$ for some $\lambda\in\mathbb{C}$.

This is a version of Shur's Lemma using the language of intertwiners.

1.1.1 Intertwiners of Lie Algebras

Now let $\mathfrak{g}$ be a Lie algebra and let (V,ρ), (W,σ) be representations of $\mathfrak{g}$, i.e., Lie algebra homomorphisms

$$\rho:\mathfrak{g}\to\mathfrak{gl}(V),\quad \sigma:\mathfrak{g}\to\mathfrak{gl}(W). \tag{4}$$

A **intertwiner of Lie algebra representations** is a linear map $\varphi:V\to W$ satisfying

$$\varphi\circ\rho(X)=\sigma(X)\circ\varphi,\quad \forall X\in\mathfrak{g}. \tag{5}$$

Equivalently, φ intertwines the $\mathfrak{g}$-actions:

$$\varphi(X\cdot v)=X\cdot\varphi(v),\quad \forall X\in\mathfrak{g},,v\in V, \tag{6}$$

where $X\cdot v:=\rho(X)(v)$.

The category of representations of $\mathfrak{g}$ with intertwiners as morphisms is denoted $\mathrm{Rep}(\mathfrak{g})$.

Example

Let $\mathfrak{g}=\mathfrak{sl}_2(\mathbb{C})$ with standard generators e,f,h satisfying $[h,e]=2e$, $[h,f]=-2f$, $[e,f]=h$. The inite-dimensional irreducible representations V_n $(n\in\mathbb{N})$ have basis $\{v_0,v_1,...,v_n\}$ with

$$h\cdot v_k=(n-2k)v_k,\quad e\cdot v_k=(n-k+1)v_{k-1},\quad f\cdot v_k=(k+1)v_{k+1}. \tag{7}$$

By Schur's Lemma, $\mathrm{Hom}_{\mathfrak{sl}_2}(V_m,V_n)=0$ for $m\neq n$, and $\mathrm{End}_{\mathfrak{sl}_2}(V_n)\simeq\mathbb{C}$.

1.1.2 CVO as Intertwiners

The Chiral Vertex Operators constructed in Moore and Seiberg's work can be understood as king of a generalization of the concept of intertwiners of representation of Virasoro Algebra (in fact any Chiral Algebra(eg. Kac Moody Algebra) can be used here).

See [2] for a constructive definition of CVOs. In the constructive definition, we are made believe that the CVOs are defined to compute the BPZ conformal blocks of RCFT (at least for ones on a plane and on a torus).

There is also an axiomatic definition of CVOs as a intertwining object between:

$$\binom{i}{jk}(z):\mathcal{A}\otimes\mathcal{A}\to\mathcal{A} \tag{8}$$

where $\mathcal{A}$ is a representation of the chiral algebra of CFT, eg. Virasoro Algebra. However, instead of following Equation 6 as a normal lie algebra, it satisfy a more complicated intertwining relation:

We frist define a "co-product" of the chiral algebra representation:

$$\begin{aligned}&\Delta_z : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A},\\ &\Delta_z(L_n) = 1 \otimes L_n + \sum_{k=0}^{\infty} \binom{n+1}{k} z^{n+1-k} L_{k-1} \otimes 1.\end{aligned} \tag{9}$$

This kinds of bizzare co-product in fact follows directly from Cauchy theorem of complec integral on with 3 point instertion of the virasoro generators.

Then we define CVO of intertwining objects satisfying:

1. **Intetwining** compatible to the co-product:

$$\begin{aligned}(L_n)\binom{i}{jk}(z)(\beta \otimes \gamma) &= \binom{i}{jk}(z)(\Delta_z(L_n)(\beta \otimes \gamma))\\ &= \binom{i}{jk}(z)\left[(\beta \otimes L_n\gamma) + \sum_{k=0}^{\infty}\binom{n+1}{k} z^{n+1-k}(L_{k-1}\beta \otimes \gamma)\right]\end{aligned} \tag{10}$$

1. **Position dependence**:

$$\partial_z \binom{i}{jk}(z)(\beta \otimes \gamma) = \binom{i}{jk}(z)(L_{-1}\beta \otimes \gamma) \tag{11}$$

This abstract definition of CVO is equivalent to the constructive definition;

$$\binom{i}{jk}(z)(\beta \otimes *) = \Phi_{ik}^{j,\beta}(z) \tag{12}$$

where we view $\Phi_{ik}^{j,\beta}(z)$ as kinds of a chiral primary field with insertion at point z with a field in j representation and labeled by β.

To understand the mathematical structure of CFT, we commonly say call the space of CVOs with fixed i, j, k representation as V_{jk}^i and this is in fact by this definition:

- Mathematically, V_{jk}^i is the space of intertwiners of representations of chiral algebra with the above intertwining relation.
- Semiphysically, V_{jk}^i is the space of chiral primary fields with fixed representation i, j, k and the above position dependence and conformal transformation property.
- Physically, V_{jk}^i is the space of all possible conformal blocks with fixed representation i, j, k and position $0, z, \infty$.

Naively, we may think that the space V_{jk}^i is a 1 dimensional vector space due to the fact that the 3 point blocks is fully fixed by conformal symmetry. This is true for Virasoro Minimal Models, yet for general Kac Moody Theory, this space may be of higher dimension.

We often notation $N_{jk}^i := \dim V_{jk}^i$ as the fusion rule of the CFT, which counts the number of independent conformal blocks with fixed representation i, j, k and position $0, z, \infty$.

1.2 Vertex Operator Algebras

We know that a CFT should have a chiral algebra, well at least the Virasoro Algebra. You do can add extra symmetries and have more, like Kac Moody Algebra … And we know that

the Hilbert Space of the CFT should form a direct sum of unitary representation of the chiral algebra:

$$\mathcal{H} = \bigoplus_i \mathcal{H}_i, \quad \mathcal{H}_i = V_i \otimes \overline{V}_i, \tag{13}$$

where V_i is the representation of the chiral algebra. In simple cases like the Virasoro Minimal Models, the representation of Virasoro Algebra may quite uniquely describe the CFT.

However, in more general and complicated cases, a same Virasoro Algebra (with definite central charge) may have many different CFT theories. A fundamental example is the $c = 1$ Virasoro Algebra, which can give rise to tons of models.

Thus, one may want to have a more delicate and precise characterization of the theory, this is where the Vertex Operator Algebra comes in. However, for our understanding one may imagine it as the Virasoro Algebra and its representation as the representation of the Virasoro Algebra, as Moore and Seiberg originally did in their work.

1.2.1 Module and Intertwiner

For CFT what is important is the representation and the intertwiners of VOA. One may analogue it as a generalization:

- Representation of Chiral Algebra (Virasoro, Kac Moody, ...) $\sim$ Representation of VOA (often called a module of VOA)
- Chiral Vertex Operator $\sim$ Intertwiner of VOA modules

With these preparation we can see that there is a natural mathematical structure:

- We can view Representations as kinds of **objects**
- And we view the intertwiners as kinds of **morphisms** between these objects.

This is how a Modular Tensor Category is born in CFT and which will be the main topic of the next section.

2 Modular Tensor Categories

We already see that CFT conformal blocks give rise to defining representations and CVOs and their structure naturally forms a **object** and **morphism** structure, which is the basic ingredient of a Category. Thus, we can use the language of Category to describe the structure of CFT, and this is what we call a Modular Tensor Category.

2.1 Category Basics

A Category $\mathcal{C}$ is a mathematical structure with 2 typse of data:

- Objects: $\mathrm{Obj}(\mathcal{C})$, for example, $U, V \in \mathrm{Obj}(\mathcal{C})$.
- Morphisms: for any two objects $U, V \in \mathrm{Obj}(\mathcal{C})$, there is a set of morphisms $\mathrm{Hom}(U, V)$, for example, $f \in \mathrm{Hom}\ (U, V)$.

We have much to say about some basic properties we assume:

1. **Composition** of morphisms: for any $f \in \mathrm{Hom}(U, V)$ and $g \in \mathrm{Hom}(V, W)$, there is a morphism $g \circ f \in \mathrm{Hom}(U, W)$.
2. **Identity Exist**: for any $U \in \mathrm{Obj}(\mathcal{C})$, there is a special morphism $\mathrm{id}_U \in \mathrm{Hom}(U, U)$, called the **identity morphism** of U, it satisfies:

$$\mathrm{id}_V \circ f = f = f \circ \mathrm{id}_U, \quad \forall f \in \mathrm{Hom}(U, V). \tag{14}$$

1. **Completeness** The set $\mathrm{Obj}(\mathcal{C})$ is complete under direct sum, i.e., for any $U, V \in \mathrm{Obj}(\mathcal{C})$, there is an object $U \oplus V \in \mathrm{Obj}(\mathcal{C})$.

Under this general definition for a quite general Category, we now can introduce more and more structures on it to finally construct a Modular Tensor Category, which is extensively used in the study of Conformal Field Theory and Topological Quantum Field Theory.

2.2 Modular Tensor Category

2.2.1 Abelian Category

An **Abelian Category** means:

1. There is a 0 Object $0 \in \mathrm{Obj}(\mathcal{C})$;
2. Morphisms possess many properties as follows:

- Every morphism set $\mathrm{Hom}(A, B)$ is an **abelian group**, that means that every morphism have structure of:
  - addition: for any $f, g \in \mathrm{Hom}(A, B)$, there is a morphism $f + g \in \mathrm{Hom}(A, B)$;
  - zero morphism: there is a morphism $0 \in \mathrm{Hom}(A, B)$ such that $f + 0 = f$ for all $f \in \mathrm{Hom}(A, B)$;
  - additive inverse: for any $f \in \mathrm{Hom}(A, B)$, there is a morphism $-f \in \mathrm{Hom}(A, B)$ such that $f + (-f) = 0$.
  - Abelian property: for any $f, g, h \in \mathrm{Hom}(A, B)$, we have $f + (g + h) = (f + g) + h$ and $f + g = g + f$.
- composition of morphisms is bilinear;

$$\begin{gathered} f \circ (g + h) = f \circ g + f \circ h, \quad (f + g) \circ h = f \circ h + g \circ h, \\ \forall f, g \in \mathrm{Hom}(B, C), \forall h \in \mathrm{Hom}(A, B). \end{gathered} \tag{15}$$

- Every morphism has a **kernel** and a **cokernel**

Note **Kernel and Cokernel**

Given a morphism $f : A \to B$ in $\mathcal{C}$:

- The **kernel** of f is an object K together with a morphism $\iota : K \to A$:

$$\ker(f) = (K, \iota) \tag{16}$$

such that $f \circ \iota = 0$ (0 is assume by the abelian structure), and which is universal with this property: for any Z and any $\varphi : Z \to A$ with $f \circ \varphi = 0$, there exists a unique morphism $u : Z \to K$ with $\iota \circ u = \varphi$.

- The **cokernel** of f is an object Q together with a morphism $\pi : B \to Q$:

$$\mathrm{coker}(f) = (Q, \pi) \tag{17}$$

such that $\pi \circ f = 0$, and which is universal: for any Z and any $\psi : B \to Z$ with $\psi \circ f = 0$, there exists a unique $v : Q \to Z$ with $v \circ \pi = \psi$.

- Every monomorphism is the (morphism part of) kernel of its cokernel, and every epimorphism is the cokernel of its kernel:

$$h = \ker(\mathrm{coker}(h)) \quad g = \mathrm{coker}(\ker(g)) \tag{18}$$

This means monomorphisms and epimorphisms are "as well-behaved as injections and surjections of vector spaces."

Note **Monomorphisms and Epimorphisms**

A morphism $h : A \to B$ is a **monomorphism** if $h \circ \alpha = h \circ \beta$ implies $\alpha = \beta$ (left-cancellable), and an **epimorphism** if $\alpha \circ h = \beta \circ h$ implies $\alpha = \beta$ (right-cancellable).

- Every morphism f admits a factorization $f = h \circ g$, where h is a monomorphism and g is an epimorphism.

The above are tedious and boring mathematical construction, for physicist is better to understand as:

- Morphisms are sort of "Linear maps" between objects, and the composition is just the composition of linear maps.

2.2.2 Tensor Category

A tensor category means that there is a tensor product $\otimes$ on the category defined both for objects and morphisms, such that:

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Bibliography

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[2] G. W. Moore and N. Seiberg, "Lectures on RCFT," in *1989 Banff NATO ASI: Physics, Geometry and Topology*, Sep. 1989, pp. 1–129.